CH 1 — Abstract Vector Spaces

Luke Lu • 2026-10-19

1.1 Vector Spaces

Ideas: $\mathbb{R}^n$ is useful. There are other objects that "behave" and "look" like $\mathbb{R}^n$

Examples:

1. $V = P_n(\mathbb{F}) = \{a_0 + a_1x + a_2x^2 + ... + a_nx^n : a_i \in \mathbb{F}\}$

   Polynomial degree $\leq n$:

   Additive property

   $$a_0 + a_1x + + ... + a_nx^n + b_0 + b_1x + + ... + b_nx^n = (a_0 + b_0) + (a_1 + b_1)x + ... + (a_n + b_n)x^n$$

   Scalar Multiplicative property

   $$c(a_0 + a_1x + ... + a_nx^n) = ca_0 + ca_1x + ... + ca_nx^n$$

2. $V = M_{m\times n}(\mathbb{F}) = \left\{ \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \cdots & \cdots & \cdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} : a_{ij} \in \mathbb{F} \right\}$

   Same properties for matrices like the above

   Observation:

$\mathbb{R}^4$	$P_3(\mathbb{R})$	$M_{2\times2}(\mathbb{R})$
$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$	$a + bx + cx^2 + dx^3$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

   Notice that the above not only "look alike", they share similar operations (addition and scalar multiplication)

   This is isomorphism

3. $V = F(\mathbb{R}, \mathbb{R}) = \{\text{functions}, f : \mathbb{R} \to \mathbb{R}\}$

   We can add and scalar multiply

   Question: Does $F(\mathbb{R}, \mathbb{R})$ is isomorphic to $\mathbb{R}^n$

   ANS: No, the reason will be learned later in the course. The current explanation is that $F(\mathbb{R}, \mathbb{R})$ is an infinity-demensioned vector space.

📘 **Definition — Vector Space**

A vector space over $\mathbb{F}$ is a set V together with two operations $\begin{cases}\text{addition}(+)\\ \text{scalar multiplication}(\cdot)\end{cases}$ s.t.

$$\forall \vec{x}, \vec{y},\ \vec{x}+\vec{y} \subseteq V$$

$$\forall c \in \mathbb{F}, c\vec{x} \subseteq V$$

Furthermore, the following conditions must hold for $\forall \vec{x}, \vec{y}, \vec{z} \in V, c, t \in \mathbb{F}$

1. $\vec{x}+\vec{y}=\vec{y}+\vec{x}$
2. $(\vec{x}+\vec{y})+\vec{z}=\vec{x}+(\vec{y}+\vec{z})$
3. $c(\vec{x}+\vec{y})=c\vec{x}+c\vec{y}$
4. $c(t\vec{x})=ct(\vec{x})$
5. $(s+t)\cdot\vec{x}=s\cdot\vec{x}+t\cdot\vec{x}$
6. $1\cdot\vec{x}=\vec{x}$
7. $\exists\vec{0}\in V$ s.t. $\vec{x}+\vec{0}=\vec{x}$
8. $\forall\vec{x}\in V, \exists -\vec{x}\in V$ s.t. $\vec{x}+(-\vec{x})=\vec{0}$

⚠️ **Warning — Zero Vector** and **Additive Inverse**

Zero vector $\vec{0}$ is not the vector with every entry being 0. It is subject to change with its corresponding vector space

Additive Inverse $-\vec{x}$ are not the vector with every entry being the negation of its origin entry. Refer to the Axiom to determine the corresponding additive inverse

Both $\vec{0}$ and $-\vec{x}$ are **unique** in their corresponding vector space

Note:

1. The elements in V are called vectors
2. We omit the "$\cdot$" in $s\cdot\vec{x}$ to $s\vec{x}$
3. We write $\vec{x}-\vec{y}$ instead of $\vec{x}+(-\vec{y})$
4. $\vec{0}$ is called the **zero vector**
5. $-\vec{x}$ is called the **additive inverse** of $\vec{x}$

Examples:

1. $V=\mathbb{F}^n=\left\{\begin{bmatrix}x_1\\ \dots\\ x_n\end{bmatrix}:x_i\in\mathbb{F}\right\}, V=M_{m\times n}(\mathbb{F})=\left\{\begin{bmatrix}a_{11} & \dots & a_{1n}\\ \dots & \dots & \dots\\ a_{m1} & \dots & a_{mn}\end{bmatrix}:a_{ij}.\in\mathbb{F}\right\}$,
   and $V=P_n(\mathbb{F})=\{a_0+a_1x++\dots+a_nx^n:a_i\in\mathbb{F}\}$ with addition and scalar multiplication be usual operations.
   Here, $\vec{0}=\begin{bmatrix}0\\ \dots\\ 0\end{bmatrix}\begin{bmatrix}0 & \dots & 0\\ \dots & \dots & \dots\\ 0 & \dots & 0\end{bmatrix}, 0+0x+\dots+0x^n$
2. $V=F(\mathbb{R},\mathbb{R})=\{\text{functions}, f:\mathbb{R}\to\mathbb{R}\}$ with addition and scalar multiplication be usual operations.
   Here, $\vec{0}$ =the zero function, i.e. $z(x)=0$

3. $V = \mathbb{R}^2$

   Define addition and scalar multiplication by

   $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1+y_2 \\ x_2+y_1 \end{bmatrix}$

   $s\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} sx_2 \\ sx_1 \end{bmatrix}$

   This is NOT a vector space.

   $1\vec{x} \neq \vec{x} \implies \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \neq \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}$

   $\vec{x} + \vec{y} \neq \vec{y} + \vec{x}$

4. $V = \mathbb{R}^2$

   Define addition and scalar multiplication by

   $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1+y_1-1 \\ x_2+y_2+1 \end{bmatrix}$

   $s\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} sx_1+s-1 \\ sx_2-s+1 \end{bmatrix}$

   Check axiom 7 and 8 as 1-6 are easy and straightforward

   Axiom 7:

   Note that in this case $\begin{bmatrix} 0 \\ ... \\ 0 \end{bmatrix}$ is not the zero vector for this vector space!

   We suppose $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$,

   then $\vec{x} + \vec{0} = \vec{x} \iff \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \iff \begin{bmatrix} x_1+z_1-1 \\ x_2+z_2+1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \implies \vec{z} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \in V$

   Axiom 8:

   We suppose $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$

   then $\vec{x} + \vec{z} = \vec{0} \iff \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \iff \begin{bmatrix} x_1+z_1-1 \\ x_2+z_2+1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \implies \vec{z} = \begin{bmatrix} x_1+2 \\ x_2-2 \end{bmatrix}$

   $-\vec{x} = \begin{bmatrix} -x_1-2 \\ -x_2+2 \end{bmatrix}$

5. Proof uniqueness of $\vec{0}$

   Let V be a vector space. Suppose $\vec{0}_1, \vec{0}_2 \in V$ satisfy

   $\vec{0}_1 + \vec{x} = \vec{x}, \vec{x} \in V$

   $\vec{0}_2 + \vec{x} = \vec{x}, \vec{x} \in V$

   We want $\vec{0}_1 = \vec{0}_2$

   Proof

   $\vec{0}_2 + \vec{x} + \vec{0}_1 = \vec{x} \implies \vec{0}_2 + \vec{0}_1 = \vec{0}_1$

   $\vec{0}_1 + \vec{x} + \vec{0}_2 = \vec{x} \implies \vec{0}_1 + \vec{0}_2 = \vec{0}_2$

   Together, $\vec{0}_1 + \vec{0}_2 = \vec{0}_2 + \vec{0}_1 = \vec{0}_1 = \vec{0}_2$

□

6. Proof $0\vec{x} = \vec{0}, \forall \vec{x} \in V$

   $0\vec{x} = (0+0)\vec{x} = 0\vec{x} + 0\vec{x} \implies \vec{0} = 0\vec{x}$

1.2 Subspaces and Spans

> 📘 **Definition — Subspace**
>
> Let V be a vector space over $\mathbb{F}$ and $W \subseteq V$. We say **subspace** of V if W is also a vector space over $\mathbb{F}$ with same addition and scalar multiplication as for V

> 🐳 **Info — Subspace Test**
> Let V be a vector space over $\mathbb{F}$. Let $W \subseteq V$.
>
> Then W is a subspace of V if and only if:
>
> 1. $\vec{0} \in W$
> 2. W is closed under addition (if $\vec{x}, \vec{y} \in W, \vec{x} + \vec{y} \in W$)
> 3. W is closed under scalar multiplicaiton(if $\vec{x} \in W, c \in \mathbb{F}, c\vec{x} \in W$)

Example:

1. Let $W = \{p(x) \in P_0(\mathbb{R}) : p'(0) = 0\}$

   W is a vector space with usual $+$ and $\cdot$ of polynomials

   Most of the axioms are obvious for all polynomials.

   We need to check axioms that are less intuitive:

   1. Is the zero vector in W?
   2. Is W closed under addition?
   3. Is W closed under scalar multiplication?

   Proof

   Let $V \in P_n(\mathbb{R})$

   1. Is the zero vector in W?

      $0 + 0x + ... + 0x^n = z(\vec{x})$

      Clearly, $z'(0) = 0$ So $z(\vec{x}) \in W$

   2. Is W closed under addition?

      Let $p(x), q(x) \in W$

      $(p+q)'(0) = p'(0) + q'(0) = 0 \in W$

3. Is W closed under scalar multiplication?

   Let $p(x) \in W, s \in \mathbb{R}$.

   $(sp)'(0) = s(p'(0)) = s0 = 0 \in W$

□

2. Let V be any vector space.

   $W = V$ is a subspace. $W\left\{\vec{0}\right\}$ is a subspace.

3. $V = \mathbb{F}^n$

   The origin is a subspace.
   The lines pass through origin are subspace.
   The planes pass through origin are subspace.

4. $V = P_3(\mathbb{F}), W = \left\{a + bx + cx^2 + dx^3 : b \geq 0\right\}$

   W is not a subspace: not closed under scalar multiplicaiton.

   $2x \in W, -1(2x) \notin W$

4. $V = P_n(\mathbb{F}), W = \{p(x) \in V : p''(1) + p'(2) = 2\}$

   W is not a subspace since $\vec{0} \notin W$

Note:

- A tip is that **non-homogeneous and equations** are usually **not** a subspace
- **Linear homogeneous** equations are usually a subspace

> 📘 **Definition — Span**
>
> Let V be a vector space. Let $\vec{v}_1, \vec{v}_2, ..., \vec{v}_k \in V$.
>
> Then the **span** of $\vec{v}_1, ..., \vec{v}_k$ is the set
>
> $$W = \text{span}\{\vec{v}_1, ..., \vec{v}_k\} = \{a_1\vec{v}_1 + ... + a_k\vec{v}_k : a_i \in \mathbb{F}\}$$
>
> **Theorem**
>
> Let V be a vector space, let $\vec{v}_1, ..., \vec{v}_k$ then $W = \text{span}\{\vec{v}_1, ..., \vec{v}_k\}$ is a subspace of V

Note: the example (1) below gives a second proof that $\{A \in M_{2\times 2} : \text{tr}(A) = 0\}$ is a subspace of $M_{2\times 2}(\mathbb{F})$

Example:

1. $V = M_{n\times n}(\mathbb{F}), W = \{A \in M_{n\times n} : \text{trace}(A) = 0\}$

   Recall that $\text{trace}(A)$ is the sum of the diagonal (i.e. $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies \text{trace}(A) = a + d$)

   Claim W is a subspace of V

   Recall that

   - $\text{trace}(A + B) = \text{trace}(A) + \text{trace}(B)$
   - $\text{trace}(cA) = c\ \text{trace}(A), c \in \mathbb{R}$

Proof

1. $\vec{0} = \begin{bmatrix} 0 & ... & 0 \\ ... & ... & ... \\ 0 & ... & 0 \end{bmatrix}, \text{trace}\left(\vec{0}\right) = 0 + ... + 0 = 0 \Longrightarrow \vec{0} \in W$
2. Let $A, B \in W, c \in \mathbb{F}$.

   $\text{trace}(cA + B) = \text{trace}(cA) + \text{trace}(B) = c\ \text{trace}(A) + \text{trace}(B) = 0$ since $A, B \in W$

   Thus $cA + B \in W$, closed by addition and scalar multiplication.

□

Take $W = \{A \in M_{2\times 2}(\mathbb{F}) : \text{trace}(A) = 0\}$

For an arbitrary $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in W$

$\iff \text{trace}(A) = 0 \iff a + d = 0 \iff d = -a \iff A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & -a \end{bmatrix} (a, b, c \in \mathbb{F})$

So $W = \left\{ \begin{bmatrix} a & b \\ c & -a \end{bmatrix} : a, b, c \in \mathbb{F} \right\} = \left\{ \begin{bmatrix} a & 0 \\ 0 & -a \end{bmatrix} + \begin{bmatrix} o & b \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ c & 0 \end{bmatrix} : a, b, c \in \mathbb{F} \right\}$

$\Longrightarrow W = \text{span}\left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$

We can conclude that most of the time **imposing condition decreases degrees of freedom**

Note: differentiation and equation are **linear** operations

2. $V = P_2(\mathbb{R}), W = \{p(x) \in V : p'(1) = p(-1)\}$

   For W, an arbitrary $p(x) = a + bx + cx^2 \in W$

   $p'(x) = b + 2cx \Longrightarrow p'(1) = b + 2c, p(-1) = a - b + c$

   Then $p'(1) = p(1) \iff b + 2c = a - b + c \iff a - 2b + c = 0$

   $\iff a = 2b + c \iff p(x) = a + bx + cx^2 = (2b + c)^2 + bx + cx^2$

   Now $W = \left\{(2b + c) + bx + cx^2 : b, c \in \mathbb{F}\right\} = \left\{b(2 + x) + c(1 + x^2) : b, c \in \mathbb{F}\right\}$

   So $W = \text{span}\left\{2 + x, 1 + x^2\right\}$

   Note:

   1. Writing a subspace W in the form $W = \text{span}\{\vec{v}_1, ..., \vec{v}_k\}$ is that it is easier to write down vectors in W
   2. Having $W = \text{span}\{\vec{v}_1, ..., \vec{v}_k\}$ allows us to detect the $\dim W$. Here is a catch, some spanning set are "better" than others

      (i.e. $W = \text{span}\left\{2 + x, 1 + x^2\right\}$, instead someone came up with

      $T = \text{span}\left\{2 + x, 1 + x^2, 4 + x + 2x^2\right\} = \text{span}\{p(x), q(x), r(x)\}$)

      There is some linear relations between vectors in T:

      $r(x) = p(x) + 2q(x), p(x) = r(x) - 2q(x), q(x) = \frac{1}{2}r(x) - \frac{1}{2}p(x)$

      $\Longrightarrow p(x) + 2q(x) + (-1)r(x) = \vec{0}$

1.3 Linear Independence

📘 **Definition — Linear Independence**

Let $\vec{v}_1, ..., \vec{v}_k \in V$.

1. $\vec{v}_1, ..., \vec{v}_k$ are **linearly dependent** if $\vec{0} = c_1\vec{v}_1 + ... + c_k\vec{v}_k, c_i \neq 0$
2. $\vec{v}_1, ..., \vec{v}_k$ are **linearly independent** if they are not linearly dependent.

Theorem:

1. $\vec{v}_1, ..., \vec{v}_k$ are **linearly dependent** if and only if some v_i is a linear combination of the others.
2. $\vec{v}_1, ..., \vec{v}_k$ are **linearly independent** if and only if $c_1\vec{v}_1 + ... + c_k\vec{v}_k = \vec{0} \implies c_1 = ... = c_k = 0$)

Examples:

1. $2 + x, 1 + x^2$ are linearly independent in $P_2(\mathbb{F})$ from the previous example.

   Two vectors $\vec{v}_1, \vec{v}_2$ are linearly dependent if and only if they are a scalar multiple of the others. So they are not linearly dependent.

2. Let $S = \{1 - x, 1 + x, 1 - x^2, 1 + x^2\} \subseteq P_2(\mathbb{R})$, is S linearly independent?

   Informally, $\dim P_2(\mathbb{R}) = 3$, but S contains 4. Thus there are too many to be linearly independent.

   ANS:

   Consider the equation $c_1(1 - x) + c_2(1 + x) + c_3(1 - x^2) + c_4(1 + x^2) = \vec{0}$

   We want to solve c_1, c_2, c_3, c_4 and hope to get $c_1 = c_2 = c_3 = c_4 = 0$ is the only solution.

   $(c_1 + c_2 + c_3 + c_4) + (-c_1 + c_2)x + (-c_3 + c_4)x^2 = 0 + 0x + 0x^2$

   $$\begin{cases} c_1 + c_2 + c_3 + c_4 = 0 \\ -c_1 + c_2 = 0 \\ -c_3 + c_4 = 0 \end{cases} \implies \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 \end{array}\right]$$

   There will be a nontrivial solution if and only if $\text{null}(A) \neq 0$

   Recall the Rank-Nullity Theorem: $n = \text{rank}(A) + \text{null}(A)$

   Notice that $\text{rank}(A) \leq 3$, therefore, $\text{null}(A) = 1 \neq 0$

   Since the $\text{null}(A) \geq 0$, there exists nontrivial solution. Thus S is linearly dependent.

3. $T = \{1 + x, 1 + x^2\}$ Does T span $P_2(\mathbb{R})$. That is, is $\text{span}(T) = P_2(\mathbb{R})$?

   We expect the answer to be no. There are only 2 degrees of freedom for T. But, $\dim P_2(\mathbb{R}) = 3$. That is T is lacking one element to span a 3-dimensional space

   To show that span $T = P_2(\mathbb{R})$. We need to show $\text{span}T \subseteq P_2(\mathbb{R})$ and $P_2(\mathbb{R}) \subseteq \text{span}T$.

   Take $p(x) = a + bx + cx^2 \in P_2(\mathbb{R})$.

   $p(x) \in \text{span } T \iff p(x) = c_1(1 + x) + c_2(1 + x^2),\ c_1, c_2 \in \mathbb{R}$

   $a + bx + cx^2 = (c_1 + c_2) + c_1x + c_2x^2$

$$\begin{cases} c_1 + c_2 = a \\ c_1 = b \\ c_2 = c \end{cases} \implies \left[\begin{array}{cc|c} 1 & 1 & a \\ 1 & 0 & b \\ 0 & 1 & c \end{array}\right]$$

We want the system to be consistent for all choices.

This is the case if and only if rank $A = n = 3$. But notice rank $A \leq 2 < 3$

Thus there exists some a, b, c s.t. system is inconsistent.

$p(x) \notin T$, so $p_2(\mathbb{R}) \not\subseteq \text{span } T$

> 📘 **Definition — Basis**
>
> Let V be a vector space. A **basis** for V is a set $B \subseteq V$ satisfy:
>
> 1. $B \subseteq V$
> 2. span $B = V$
> 3. B is linearly independent

Note: In general $E_{ij} \in M_{m\times n}(\mathbb{F})$ has the ij-th entry be 1.

Examples:

1. $S' = \{1 + x, 1 - x, 1 - x^2\}$ is a basis for W
2. $\{\vec{e}_1, \vec{e}_2, ..., \vec{e}_n\}$ is the standard basis for $\mathbb{F}^n$
3. $\{1, x, x^2, ..., x^n\}$ is a standard basis for $P_{n(\mathbb{F})}$
4. $\left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right\} = \{E_{11}, E_{12}, E_{21}, E_{22}\}$ is a standard basis for $M_{2\times 2}(\mathbb{F})$
5. $\{E_{ij} : 1 \leq i \leq m, 1 \leq j \leq n\}$ is the standard basis for any $M_{m\times n}(\mathbb{F})$
6. In general, vector space V does not have a **standard basis**
7. Let W be the plane in $\mathbb{R}^3$ with equation $x + y + z = 0$

   W is a 2-dimensional subspace in a 3-dimensional space

   Then $B = \left\{\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}\right\} \subseteq W$ and B is linearly independent

   Let $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \in W$, then $x + y + z = 0 \implies y = -x - z$

   $$\implies \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ -x-z \\ z \end{bmatrix} = x\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + z\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \in \text{span } B$$

   Thus B is a basis for W. There are many other possible chocies for basis for W. Any two linearly independent vectors in W satisfy to be a basis.

8. $V = \{A \in M_{2\times 2}(\mathbb{F}) : \text{tr } A = 0\} \in M_{2\times 2}(\mathbb{F})$

   Find a basis for V

   V is 4-dimensional space, we suspect the basis are less than 4-dimensional due to presence of imposed linear condition.

   $B = \left\{\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}, \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\ 1 & 0\end{bmatrix}\right\}$.

   Since B is linearly independent, so it is a basis for V

9. $W = \left\{A \in M_{2\times 2}(\mathbb{F}) : A\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}A\right\}$

   This is a subspace of $M_{2\times 2}(\mathbb{F})$

   Let $A = \begin{bmatrix}a & b\\ c & d\end{bmatrix}$. Then $A \in W \iff \begin{bmatrix}a & b\\ c & d\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}a & b\\ c & d\end{bmatrix} \iff \begin{bmatrix}a & -b\\ c & -d\end{bmatrix} = \begin{bmatrix}a & 0\\ 0 & -d\end{bmatrix}$

   $\implies -b = b, -c = c \iff b = c = 0$

   $A = \begin{bmatrix}a & 0\\ 0 & d\end{bmatrix} = a\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix} + d\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}$

   Therefore $B = \left\{\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}\right\}$, and it is linearly independent, thefore, it is a basis.

10. $V = \left\{\vec{0}\right\}$, zero vector space.

    By convention, $B = \varnothing$ is a basis for V.

    $\varnothing$ is linearly independent, span $\varnothing = \left\{\vec{0}\right\}$

QUESTION: Does every vector space have a basis?

ANSWER: Yes but for finite-dimensional vector space for this course.

(i.e. This is infinity-dimensional: $V = F(\mathbb{R}, \mathbb{R}) = \{\text{functions}, f : \mathbb{R} \to \mathbb{R}\}$)

> 📘 **Definition — Finite-Dimensional**
>
> A vector space is **finite-dimensional** if it has a finite spanning set: $V = \text{span}\{\vec{v}_1, ..., \vec{v}_k\}$

$M_{m\times n}(\mathbb{F})$ and $P_n(\mathbb{F})$ are finite-dimensional spaces.

Spaces of functions: $f : \mathbb{R} \to \mathbb{R}$ are infinity-dimensional spaces.

> 📘 **Definition — Vector Space Basis**
>
> Let V be a finite-dimensional vector space.
>
> 1. V has a basis
> 2. Any two bases of V have the same size. (In other words, if B and C are bases for V, then $|B| = |C|$)
> 3. Every subspace of V is finite-dimensional

Proof

1. V has a basis

   By definition, V has a finite spanning set: $S = \{\vec{v_1}, ..., \vec{v_k}\}$

   If S is linearly independent, then S is a basis.

   If not, say $\vec{v_k}$ is a linear combination of others, then remove $\vec{v}_k$ and call it $S' = \{\vec{v}_1, ..., \vec{v}_{k-1}\}$

   Now S' spans V

2. Any two bases of V have the same size

   Let B and C be two bases for V,

   if $|B| > |C|$, $|B|$ is not linearly independent.

   if $|B| < |C|$, B does not span V.

   $\implies |B| = |C|$

> 💡 **Tip — Proposition**
>
> Let V be a vector space with basis B of size n
>
> Let S be a finite subset of V
>
> - if $|S| > n$, then S is linearly independent
> - if $|S| < n$, then S cannot span V

> 📘 **Definition — Dimension**
>
> Let V be finite-dimensional. Then the **dimension** of V, $\dim V$, is the size of any basis for V

Remark: By the theorem, $\dim V$ is well-defined

Example:

V	Basis	$\dim V$
$\{\vec{0}\}$	$\varnothing$	0
$\mathbb{F}^n$	$\{\vec{e}_1, ..., \vec{e}_n\}$	n
$M_{m\times n}(\mathbb{F})$	$\{E_{11}, E_{1i}, ..., E_{ij}\}$	mn
$P_n(\mathbb{F})$	$\{1, x, ..., x^n\}$	$n+1$
Plane in $\mathbb{R}^n$ through $\vec{0}$	Any 2 LI vectors in plane	2
$\{A : M_{2\times 2}(\mathbb{F}) : \text{tr}A = 0\}$	$\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}, \begin{bmatrix}0 & 1\\ 0 & 1\end{bmatrix}, \begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}$	3
$\text{span}\{x, x+x^2, x-x^2\} \subseteq P_{10}(\mathbb{R})$	$\{x, x+x^2\}$	2
$\{p(x) \in P_3(\mathbb{R}) : p(1) = 0, p(-1) = 0\}$	$\{x^2-1, x^3-x\}$	2

Note: V in the last row is a subset of $P_3(\mathbb{R})$, not necessarily be 4 dimensional

We need to find a spanning set for $\{p(x) \in P_3(\mathbb{R}) : p(1) = 0, p(-1) = 0\}$

$p(x) = a + bx + cx^2 + dx^3 \in V \iff \begin{cases} p(1)=0 \\ p(-1)=0 \end{cases} \iff \begin{cases} a+b+c+d=0 \\ a-b+c-d=0 \end{cases}$

$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & -1 & 0 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array}\right] \implies a = -c, b = -d \implies p(x) = -c + -dx + cx^2 + dx^3$

$\implies c(x^2 - 1) + d(x^3 - x) \implies V = \text{span}\{x^2 - 1, x^3 - x\}$

Notice that $B = \{x^2 - 1, x^3 - x\}$ is a spanning set for V and is linearly independent, thus it is a basis for V

> 📘 **Definition — Dimension and Vector Space**
>
> Let V be a finite-dimensional vector space. Let $W \subseteq V$ be a subspace.
>
> 1. W is finite-dimensional
> 2. $\dim W \leq \dim V$
> 3. $\dim W = \dim V \iff W = V$

Proof

2. $\dim W \leq \dim V$

   Let B be a basis for W ($|B| = \dim W$).

   Since $B \subseteq W \subseteq V$, we have $|B|$ linearly independent vectors in V.

   But $|\text{Linearly independent}| \leq \text{dim}$, so $|B| \leq \dim V$.

   Thus $\dim W \leq \dim V$.

3. $\dim W = \dim V \iff W = V$

   $\implies$

   W have that $\dim W = \dim V$, then if B is a basis for W, then it is also a basis for V. Since B is linearly independent having $\dim V$. Thus $V = \text{span}\{B\} = W$

   Conversely also applys.

1.4 Coordinates

In $\mathbb{R}^3$, let $S = \left\{ \vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \vec{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

Then any $\vec{x} \in \mathbb{R}^3$ is a linear combination of $\vec{e}_1, \vec{e}_2, \vec{e}_3$

In $\mathbb{R}^2$, let $B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. This is a basis.

Consider $\vec{v} = \begin{bmatrix} 3 \\ 5 \end{bmatrix} \in \mathbb{R}^2$

Then $\vec{v} = 4\begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-1)\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

📘 **Definition — Coordinates**

Let $B = \left\{\vec{b}_1, ..., \vec{b_n}\right\}$ be an ordered basis for V,

Then the $B-$ coordinates of $\vec{v} \in V$ are the coefficients of $\vec{v}$ when we express it as a linear combination of V

$$\vec{v} = a_1\vec{b}_1 + ... + a_n\vec{b}_n$$

The $B-$ coordinates vector of $\vec{v}$ is $[\vec{v}]_B = \begin{bmatrix} a_1 \\ ... \\ a_n \end{bmatrix}$

Examples:

1. Let $V = P_3(\mathbb{R}), B = \{1, x, x^2, x^3\}$

   Then $p(x) = 1 - x^3 = 1 + 0x + 0x^2 - x^3$
2. $V = M_{2\times 2}(\mathbb{R}), B = \left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right\}$

   $A = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix} = E_{11} + E_{12} + 0E_{21} + 3E_{22} \implies [A]_B = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 3 \end{bmatrix}$
3. $V = \{A \in M_{2\times 2}(\mathbb{F}) : \text{tr } A = 0\}\ B = \left\{\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right\}$

   $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix} = 2\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + 3\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 5\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} ==? [A]_B = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$

📘 **Definition — Uniqueness of Coordinate Vector**

Let $B = \{\vec{v}_1, ..., \vec{v}_n\}$ be a basis for V. If $\vec{v} \in V, \exists$ unique $c_1, ..., c_n \in \mathbb{F}$ s.t.

$$\vec{v} = c\vec{v}_1 + ... + c_n\vec{v}_n$$

Proof

Suppose $\vec{v} = c_1\vec{v}_1 + ... + c_n\vec{v}_n = d_1\vec{v}_1 + ... + d_n\vec{v}_n, c_i \in \mathbb{F}, d_i \in \mathbb{F}, i \in \mathbb{N}$

Then $\vec{v} - \vec{v} = \vec{0} = (c_1 - d_1)\vec{v}_1 + ... + (c_n - d_n)\vec{v}_n$

By linear independence, $\begin{cases} c_1 - d_1 = 0 \\ c_2 - d_2 = 0 \\ ... \\ c_n - d_n = 0 \end{cases} \implies c_i = d_i, \forall i$

□

📘 **Definition — Linearity of Coordinates**

Let B be a basis for V, then

1. $\left[\vec{0}\right]_B = \begin{bmatrix} 0 \\ ... \\ 0 \end{bmatrix} \in \mathbb{F}^n$
2. $[\vec{x} + \vec{y}]_B = [\vec{x}]_B + [\vec{y}])_B \in \mathbb{F}^n$
3. $[c\vec{x}]_B = c[\vec{x}]_B \in \mathbb{F}^n$

Proof

Let $B = \{\vec{v}_1, ..., \vec{v}_n\}$

1. $\vec{0} = 0\vec{v}_1 + ... + 0\vec{v}_n$

   By uniqueness, $\left[\vec{0}\right]_B = \begin{bmatrix} 0 \\ ... \\ 0 \end{bmatrix}$

2. Suppose $\vec{x} = c_1\vec{v} + ... + c_n\vec{v}_n, \vec{y} = d_1\vec{v}_1 + ... + d_n\vec{v}_n$

   Then $\vec{x} + \vec{y} = (c_1 + d_1)\vec{v}_n + ... + (c_n + d_n)\vec{v}_n$

   Thus $[\vec{x} + \vec{y}]_B = \begin{bmatrix} c_1+d_1 \\ ... \\ c_n+d_n \end{bmatrix} = \begin{bmatrix} c_1 \\ ... \\ d_1 \end{bmatrix} + \begin{bmatrix} d_1 \\ ... \\ d_n \end{bmatrix} [\vec{x}]_B + [\vec{y}])_B$

3. Similar to (2)

□

Example:

1. Let $B = \{\vec{v}_1, ..., \vec{v}_n\}$ be a basis for V. Let $S = \{\vec{x}_1, ..., \vec{x}_n\} \subseteq V$
   1. S is linearly independent in V if and only if $\left\{[\vec{x}_1]_B, ..., [\vec{v}_n]_B\right\}$ is linearly independent in $\mathbb{F}^n$
   2. S spans V if and only if $\overline{S} = \left\{[\vec{x}_1]_B, ..., [\vec{v}_n]_B\right\}$ spans $\mathbb{F}^n$

   Proof

   1. Found in solution of lecture exercise 8
   2. We want to show that span $S = V$ if and only if $\overline{S}$ span $= \mathbb{F}^n$

      $\implies$ Assume span $S = V$.

      $\overline{S} \subseteq \mathbb{F}^n$ is trivial

      Conversely, let $\begin{bmatrix} a_1 \\ ... \\ a_n \end{bmatrix} \in \mathbb{F}^n$. We want to show that $\begin{bmatrix} a_1 \\ ... \\ a_n \end{bmatrix} \in$ span $\overline{S}$

      Consider $\vec{v} = a_1\vec{v}_1 + ... + a_n\vec{v}_n \in V =$ span S, then we know that $\vec{v} = c_1\vec{x}_1 + ... + c_n\vec{x}_n$

      $[\vec{v}]_B = [c_1\vec{x}_1 + ... + c_n\vec{x}_n]_B \implies \begin{bmatrix} a_1 \\ ... \\ a_n \end{bmatrix} = c_1[\vec{x}_1]_B + ... + c_n[\vec{x}_n]_B$

      $\impliedby$ Assume $\overline{S}$ span $\mathbb{F}^n$

      We want span $S = V$

      $S \subseteq V$ as given

      Conversely, if $\vec{v} \in V$, we want $\vec{v} \in$ span S

      Consider $[\vec{v}]_B \in \mathbb{F}^n =$ span $\overline{S}$

      $[\vec{v}]_B = c_1[\vec{x}_1]_B + ... + c_n[\vec{x}_n]_B = [c_1\vec{x}_1 + ... + c_n\vec{x}_n]_B$ (By linearity)

      $\vec{v} = c_1\vec{x}_1 + ... + c_n\vec{x}_n$ (By uniqueness of coordinates)

      So $\vec{v} \in \text{span}\{\vec{x}_1, ..., \vec{x}_n\}$

2. Show $B = \{-x + 2x + 2x^2, 2 + x^2, -3 + x\}$ is a basis for $P_2(\mathbb{R})$

   Consider $\overline{B} = \left\{ \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$

   We want to show that $\overline{B}$ is linearly independent as it has 3 elements in a 3 dimensional space.

   Suppose dim $V = n$ and let $S \subseteq V$ have $|S| = n$.

   Then S is linearly independent $\iff$ S span $V \iff$ S is basis for V

   Find $\left[1 + x + x^2\right]_B$